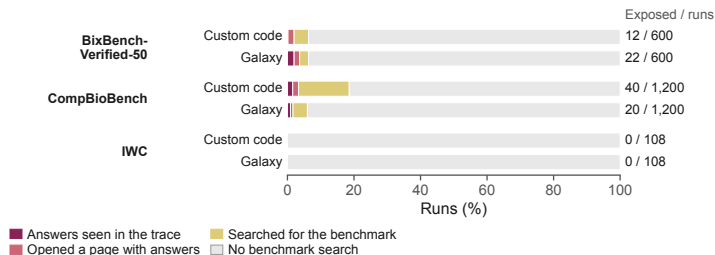


a Runs that reached benchmark during the run

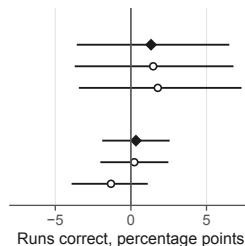
Highest tier per run, from the traces (web pages themselves are not logged)

**b** Galaxy minus custom code without exposed runs**BixBench-Verified-50**

All runs (1,200)
 Without verified or probable exposure (1,166)
 Without any benchmark search (1,124)

CompBioBench

All runs (2,400)
 Without verified or probable exposure (2,340)
 Without any benchmark search (2,105)

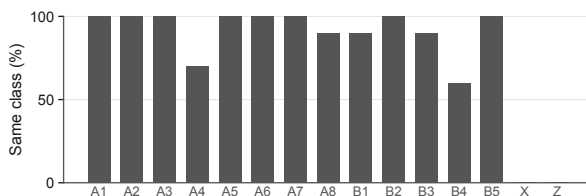
**c** Second rater for the failure-cause audit (BixBench-Verified-50)

45 incorrect runs; agreement 78%, Cohen's $\kappa = 0.56$; 96% counting secondary causes

Original audit	Validation	3	3	2	0	0
	Benchmark	3	27	0	0	0
	Knowledge	0	1	2	0	0
	Galaxy	1	0	0	1	0
	No answer	0	0	0	0	2
		Validation	Benchmark	Knowledge	Galaxy	No answer
		Second rater				

d Second rater for the failure classes of failed requests

150 requests, 10 per class; agreement 92% for classified requests ($\kappa = 0.92$); the rule's improvement group was among the rater's in 71%



Rater judged the failure an agent error only in 3%; more than one improvement plausible in 35%